

□ 05 Probability models – the normal / Gaussian distribution

Consider each of the following quantities.

- (a) The time a student takes to walk to college each morning
- (b) The playing time provided by one battery charge for an MP3 player
- (c) The area of a fence which can be covered by a 5-litre bottle of spray paint
- (d) The actual amount of orange juice in a “1 litre” carton
- (e) The length of string in a ball nominally sold in 30m lengths
- (f) The weights of a random group of adult Irish men.

A number of characteristics are shared by all.

A degree of random variation is evident in all cases. For example, this morning’s walk to college might have taken 14 minutes 20 seconds. Yesterday it might have been 13 minutes 50 seconds. Or two balls of “30 m” string are very likely to have at least some difference in their actual length.

The quantities are continuous in nature. There is no limit on the number of different volumes that an orange juice carton might contain.

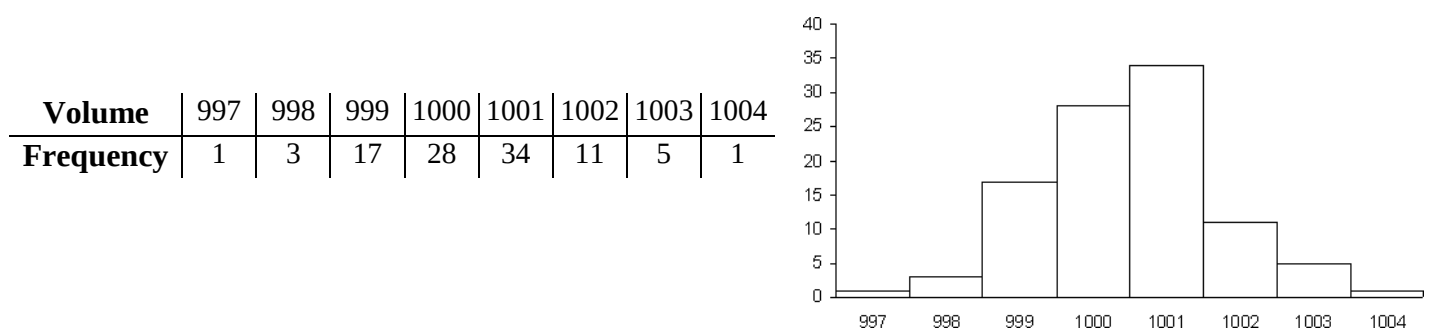
Equivalently any two volume measures however close always contain another volume measure between them.

E.g., the measures 998 ml and 998.001 ml contain the measure 998.0005. (We may not be able to detect the difference between 998 ml and 998.001 in practice, but this is a limitation of our measuring techniques).

Such (random) continuous quantities differ fundamentally from the quantities we modelled using the binomial and the Poisson which were discrete in nature.

E.g., We could consider the number of defects in a sample of 10 being 2 or 3 but not 2.5!

In addition, each of the quantities above are likely to share three other characteristics. To illustrate these, we will consider the orange juice carton example in more detail. Suppose we were to measure the volume of a batch of 1,000 “1 litre” orange juice cartons, tabulate our results (to the nearest 1 mm), compile the frequency table and construct the corresponding histogram. We might expect to get something like the following.

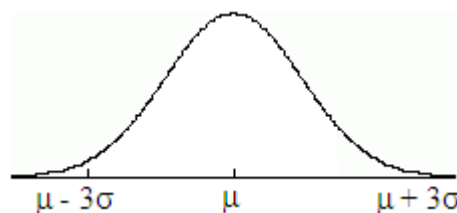


We can identify 3 characteristics of this histogram and deduce 3 corresponding characteristics in what we are measuring.

1. The more central histogram bars are taller.
Deduce: Most observations are concentrated about some central (mean) value.
2. The histogram is roughly symmetric about a vertical line through its centre.
Deduce: Observations in an interval a given distance above the mean have the same frequency as observations in a similar interval below the mean.
3. The histogram “tails off” rapidly for intervals a long way from the mean.
Deduce: Observations a long way from the mean are comparatively unlikely.

If you now reconsider each of the examples above, you may agree that all share these 3 characteristics. Such quantities are likely to have a normal or Gaussian probability distribution function.

If for any of the examples above, we to collect a very large amount of data and construct a more refined histogram we would find our graph taking a “bell” like shape as shown in the diagram. Such a normal distribution of data can be completely specified using two parameters. The mean μ and the standard deviation σ . The former is a location parameter and indicates the position of the middle of the bell. The latter is a spread parameter and indicates the “thickness” of the bell.



Once we establish that a quantity does follow a normal distribution (and as with the Poisson and other statistical models there are quantitative techniques to address this question) and once we know or can estimate the parameters μ and σ then we can perform probability calculations such as the likelihood that a randomly selected orange juice carton containing less than 1002 ml or the probability that tomorrow’s walk to college will take between $13\frac{1}{2}$ and $14\frac{1}{2}$ minutes.

Problem 1

Suppose that in our orange juice carton example the contents follow a normal distribution with mean 1000 and standard deviation 1.24 millilitres how likely is it that a randomly selected orange juice carton will have contents less than 1002 ml?

Normal tables

To use the standard normal tables, note that our Z score 1.67 can be written as $1.6 + 0.07$. Thus, we refer to the row labelled **1.6** and the column labelled **0.07** and read off the referenced entry.

A section of this table is reproduced opposite. (The full tables are available at the end of this document). The required probability measure is 0.9525 (circled in the table) or 95.25% which we can round to 95%.

Standard normal tables			
	0.06	0.07	0.08
1.5	0.9406	0.9418	0.9429
1.6	0.9515	0.9525	0.9535
1.7	0.9608	0.9616	0.9625
1.8	0.9686	0.9693	0.9699
1.9	0.9750	0.9756	0.9761

Thus 95% of OJ cartons or about 1 in 20 will have contents less than 1002 ml.

(Left tail tables are available in the department of education “log” books but omit the 0. in front of table entries. i.e., $\Pr(Z < 1.67)$ would read 9525 in those tables)

Note that for a continuous distribution like the normal and unlike discrete distributions like the Poisson and binomial we do not distinguish between $<$ and $\leq$. e.g., 1002.9999999 ... is less than but can be made arbitrarily close to 1003. For convenience we only use the relators $<$ and $>$ and dispense with $\leq$ and $\geq$. Note that it is never sensible to use ‘=’ with a continuous distribution. $\Pr(X = 1003) = \Pr(X = 1003.0000 \dots) = 0$.

Problem 2

How likely is it that a randomly selected carton has contents greater than 1001 ml?

Problem 3

How likely is it that a randomly selected carton has contents less than 998 ml?

Note that we can see that about 95% of all cartons have contents less than 2 ml above the mean of 1000 ml, i.e., 1002 ml. This of course means that 5% of cartons have contents more than 2ml above the mean. Also, 5% of cartons have contents less than 2 ml below the mean due to the symmetry of the normal distribution.

Problem 4

How likely is it that a randomly selected carton has contents more than 997 ml?

Problem 5

How likely is it that a randomly selected carton has contents between 1002 and 1004 ml?

Note

The identity used here is not a special property of the normal distribution or even of probability. It is a counting principle. E.g., to count the people in a group aged between 20 and 30 ($20 < X < 30$) one could count all those less than 30 ($X < 30$) and subtract the number less than 20 ($X < 20$).

Take care to avoid the common mistakes of writing $\Pr(A < X < B) = \Pr(X < B) - \Pr(X > A)$ (note the $>$ relator in the last term) or attempting to apply a construct of the form $1 - \text{“something or other”}$. The identity used here is the way to tackle these “sandwich” type problems)

Problem 6

How likely is it that a randomly selected carton has contents less than 1010 ml?

Problem 7

How many O.J. cartons in a batch of 420 are likely to have volume of 998 ml or less?

Problem 8

If the producers of the orange juice can sell their product at a profit of 80c per litre but cartons which contain less than 998 litres must be topped up at a cost of 3c per litre estimate the profits on a batch of 850 litres.

As only €1.21 is lost out of €680 the producers are not being seriously affected in financial terms by the production of bad cartons. Of course, the situation would be different if the mean or standard deviation were to change, a common problem in production processes.

Exercise

Which of the following scenarios would be most detrimental to profit as calculated above?

- (g) The mean contents of the cartons were to fall to 997 ml
- (h) The standard deviation were to rise to 2.5 ml
- (i) The cost of reprocessing a bad carton were to rise to 30c per carton.

Problem 9

The time taken by a truck driver to travel from Waterford to Dublin is normally distributed with mean $2\frac{3}{4}$ hours and standard deviation 11 minutes. How likely is it that one journey will take

- (a) at most 3 hours? (b) more than 2 hours 50 minutes? (c) between 2 hours 40 minutes and 3 hours 10 minutes
- (d) Estimate the driver's earning on 100 such trips if a basic pay is €60 per trip with a bonus of €20 for trips completed in $2\frac{1}{2}$ hours or less.

Exercise

Grades for a statistics exam follow a normal distribution with mean 54 and standard deviation 17.

- (a) How likely is it that one randomly selected student will achieve a distinction grade (≥ 70)?
- (b) What proportion of students will achieve a grade within 10 points of the mean?
- (c) How many students in a class of 48 might be expected to fail (< 40)?

Inverse probabilities

Rather than starting with a range of values (e.g., less than 2 hours) and calculate a corresponding probability sometimes we will start with a probability and seek the corresponding value range.

Problem 10

If 90% of OJ cartons in the earlier problem, have contents less than k what is k?

✎ Take care to avoid the common mistake of looking up 0.90 in the tables to get 0.8159. Here the requirement is to determine what Z score corresponds to a probability of 0.9 and not to determine what probability corresponds to a Z score of 0.90.

Problem 11

If 30% of OJ cartons have contents more than k what is k?

Problem 12

If 2% of cartons have contents less than k what is k

Problem 13

If 90% of cartons lie within k ml of the mean what is k?

Problem 14

For the journey time problem, determine the times k such that

- (a) 99% of all journeys take less than time k? (Solution 3 hours 10.6 min)
- (b) 80% of all journeys take at least time k? (Solution 2 hours 35.7 min)
- (c) 95% of all journeys take $\pm k$ minutes of the mean of $2\frac{3}{4}$ hours. (Solution ± 21.6 min)

Problem 15

If heights in a population are normally distributed with 10% of members have heights exceeding 1.9 m and 20% having heights below 1.5m what is the average height in this population?

Hint See up two equations in the two unknown quantities μ and σ

Normal distribution in R

If $X \sim N(1000, 1.2^2)$ then the following calls to R yield the probability of the outcome indicated

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> pnorm(998, 1000, 1.2) # Pr(X < 998)
> pnorm(998, 1000, 1.2, lower=F) # Pr(X > 998)
> pnorm(1001, 1000, 1.2) - pnorm(998, 1000, 1.2) # Pr(998 < X < 1001)
> qnorm(0.08, 1000, 1.2) # k, Pr(X < k) = 0.08
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Exercise A production facility produces a frozen food which has a mean calorie content of 130 calories per 100g. However, this varies according to a normal distribution with standard deviation 4.8 calories. If one such food product is selected at random determine the likelihood that its calorie content will be

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|--------|--|-----------|
| (i) | less than 138? | 95.2% |
| (ii) | more than 140? | 1.9% |
| (iii) | less than 128? | 33.8% |
| (iv) | more than 122? | 73.4% |
| (v) | between 133 and 145? | 26.5% |
| (vi) | between 126 and 139? | 76.7% |
| (vii) | within 6 calories of the mean? | 78.9% |
| (viii) | Find c if 95% of products have a calorie content of less than c. | 137.9 cal |
| (ix) | Find c if 25% of products have a calorie content of less than c. | 126.8 cal |
| (x) | Find k if 98% of products have a calorie content of within k of the mean | 11.2 cal |
| (xi) | Find k if 50% of products have a calorie content of within k of the mean | 3.2 cal |